criteria of 3%/3mm and 2%/2mm were obtained in the CT number comparisons, respectively. We achieved 3D Gamma passing rates of 0.999 and 0.986 with criteria of 3%/3mm and 2%/2mm in the dose comparisons, respectively. The difference of CTV D_{95} between the *sCT* and *gCT* was only 90 cGy[RBE]. The model yielded a mean SE of only 0.2 mm in the *sCT* robustness test.

Conclusion: A patient specific vision transformer-based network was developed and shown to be accurate and efficient to reconstruct 3D CT images from kV images.

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Do Spatial-Radiomics Improve Prediction of Locoregional Recurrence Following Radiotherapy for HNSCC?

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Purpose/Objective(s): Definitive chemoradiation for HNSCC has improved significantly with modern planning techniques and supportive care. However the locoregional recurrence risk (LR) persists and can range up to 30% depending on patient-specific factors. Prior research has used machine learning on radiomic features (or quantitative, sub-visual cues) from diagnostic imaging to predict LR with some success. In this study, we attempt to improve on these models by introducing 2 key innovations. First we identify "supervoxels," or sub-regions near but outside of the gross tumor volume (GTV), from which to extract radiomic features. Second we create a radiomic graph model where the supervoxels are prioritized based on their similarity to the GTV. We hypothesize that this spatial, graph-based approach can better identify regions suspicious for microscopic tumor involvement, which in turn would better predict clinical outcomes.

Materials/Methods: We identified the RADCURE (D1) and Head-Neck-Radiomics-HN1 (D2) datasets containing 2,611 patient CTs and RT structures from The Cancer Imaging Archive. D1 was divided into training, validation, and testing splits whereas D2 was used only for testing. For each patient CT, we identified 100 supervoxels within 10 voxels of the GTV. Texture-based radiomic features were then extracted from each supervoxel, and the top 20 with feature expression most similar (via Euclidean distance) to the GTV used to create a graph model. The nodes of this graph represent feature expression within the supervoxels and the edges the correlation to the GTV. Clinical features including age, sex, ECOG, chemotherapy use, tumor stage, size, and HPV status were used as a model parameter and baseline comparison. A graph attention neural network (GAT) was trained using these graphs for the LR prediction task. Comparisons were made with a traditional radiomic model and clinical features alone. Following prediction, model attention weights were extracted and used to identify which CT supervoxels were most informative to the model.

Results: Graph radiomics with clinical features resulted in AUCs of 0.834 and 0.806 for D1 and D2, respectively. Traditional radiomics with clinical features resulted in AUCs of 0.819 and 0.784 compared to clinical features alone achieving AUCs of 0.808 and 0.784. Qualitative examination of attention heatmaps revealed that our spatial radiomic model attention was heavily concentrated along cervical lymph node chains.

Conclusion: Spatial radiomics utilizing supervoxels from peritumoral areas were able to predict LR for HNSCC in large, multi-institutional datasets, outperforming other previously studied methods. It is notable that our model's performance did indeed improve on an already robust baseline for an independent test dataset, suggesting there is additional utility in our graph-based approach. Our attention maps further suggest that disease-relevant regions outside of the GTV can be identified in an unsupervised manner.

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Radiographic Predictors of Extranodal Extension on Preoperative Contrast-enhanced CT in Patients with Oropharyngeal Squamous Cell Carcinoma

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Purpose/Objective(s): Upfront treatment options for patients with oropharyngeal squamous cell carcinoma (OPSCC) include surgical resection or radiotherapy (RT). In surgically managed patients, extranodal extension (ENE) is an accepted indication for adjuvant chemoRT. Reliable methods to predict ENE using preoperative contrast-enhanced computed tomography (CECT) are needed to better select the initial treatment approach and avoid trimodality therapy.

Materials/Methods: Consecutive patients treated with surgical resection and neck dissection (ND) for OPSCC between 10/2012-10/2020 were identified. Exclusion criteria were recurrent/second primary disease, lack of preoperative CECT. Four fellowship-trained neuroradiologists reviewed preoperative CECTs and assessed the dissected neck side(s), blinded to the pathologic outcomes. Patients with bilateral ND had both sides of the neck evaluated individually. Each radiologist measured the following factors: axial and coronal/sagittal short/long axis, central necrosis, perinodal stranding, matted nodes, spiculation/irregular margins, absence of perinodal fat plane, infiltration of adjacent structures. ENE presence was based on the pathology report. Each factor was tested for association with pathologically confirmed ENE using univariate and multivariate models.

Results: In total, 162 patients with 206 neck sides were included in the analysis, 44 of which had ENE. Most patients had HPV-associated disease (90%) and tonsil/base of tongue tumors (98%). 4 neuroradiologists performed 824 individual neck assessments, 176 of which harbored ENE. On univariate analysis, all factors were significantly and consistently associated with ENE for all 4 radiologists. Various multivariate models were evaluated to best predict the risk of ENE based on these factors. The presence of matted nodes (odds ratio [OR] 6.45, $p<0.0001$) and spiculated/irregular nodal margins (OR 8.62, $p<0.0001$) were consistently associated with ENE on multiple models. After controlling for within-patient correlation, matted nodes and absence of perinodal fat plane were included in the model ($p<0.03$), as was infiltration of adjacent structures ($p=0.08$).

Conclusion: Multiple CECT nodal features associated with ENE are consistently identified by multiple radiologists. Matted nodes and spiculated/irregular nodal margins were found to be most strongly associated with ENE. These findings will guide the development of an accessible, clinically facile scoring system to predict the risk of ENE on preoperative CECT in the clinic or multidisciplinary tumor board.

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